Hacking Project:

Basic Idea:

Create a 3d “village” or city for a group of friends. Browser web app, with centralized database.

Village is started with a group photo or photo of each member of the group. Segmentation model and then a 2d photo to 3d avatar or face model is used to create a 3d avatar, and voice memos for each person or voice recordings are given to create a voice clone via eleven labs.

Each person thus will have a basic 3d avatar and voice model/voice lines in this 3d village.

Ideally there will be a procedurally generated 3d village that is consistent with house and what not, and the villagers/friends will walk around the village daily when browsing and will have some level of activity.

3D Village Ideas/Events:

Browser View of User:

- User can look at all other people in the village walking around
- Clicking on other person shows profile/feed
    - Feed is broken up into normal status updates (Part of dailies
    - Goal for Week
        - Progress Updates
    - Previous Goals
        - Previous progress updates

LLM God:

- The LLM “God” or Mayor will look through everyones updates throughout the week and in a weekly recap share what everyone did
- Tamagatchi life living the dream vibes with humorous sound effects maybe
- LLM God can be prayed to with requests for specific random events or challenges
    - If someone says the group hasn’t been having as much fun with games lately, it may decide to do fun games with the people as a challenge
    - Bribed with points/XP?

Daily Check In:

- Every day users are encouraged to check in
- Send message to people
- Update weekly goal with activity post/evidence
- Similar to Dailys in gatcha or other games
    - They gain XP or points for updating status
- With points/XP people can buy
    - Clothing items
    - Pets
    - Maybe need backyard upgrade for house before get dog
        - Not cat or fish tho
    - Pet decorations
    - Ability to upgrade house
    - Talking style changes
    - All these should probably be done with predetermined upgrades so we can vet upgrades and visuals.
- Points can be bought for SOME OBJECTS/Outfit with solana, and money will got o auditor 3rd party
- Throughout the day at random times LLM agent should decide when to send messages to people, with progress updates on other people, on the village itself, and with ability to send messages as other people of updates
    - Lance avatar saying to iraj
        - “Yo you’re fucking slacking bro, I did updates the whole week, you did none lol”

Weekly Check IN:

- Every week people are encouraged to set up a goal for the week
    - Can put money in general pot
    - Fancy math needs to be done for that
        - Only those with money in pot compete
- They are encouraged to throughout the week post status updates with photos and descriptions of it, which will be publically available to all villagers
- At end of week, auditor for the day will get every villagers goal and evidence posted, and grade them based on goal ambition, quality of progress reports, and overall work done
- Winner gets a crown, and wins pot based on fancy math
- Get badge or small star if do the goal at all
- Specific goals are useful, Scale is useful, clear progress or exceeding goals is useful, all three are needed for best points

Competition:

- Group in village can create mutually agreed upon, or global across village competition

- Solana can be entered or used for entry, with winner or winners collecting all the solana entry
- Reward could also be ability to give someone else or themselves a new outfit
    - Use computer vision here are well.
- A data entry or auditor 3rd party will be decided as either someone in the group, or not in the group
    - Data Entry will be given a small percent of solana pool for expediency, determined by village
- Users in the competition can put in entries with photo for proof to the auditor who will confirm or not and for amount of points or not
- Examples:
    - Gaming Competition
    - Study competition, whoever studies most in a week
    - Whoever plays most minecraft in a week
    - Mixed Goals
        - One person work out one hour a week
        - One person do study
        - Each person separate goal
- Avatars may send gloating imessages to people who haven't been trying
- Friendly competitions with no pot and a basic auditor who can even be participant can be made, will show up on both peoples or all participants profiles but lower stakes, or even no verification if needed

Random Events:

- An LLM will look at the village, and occasionally issue challenges or special events
    - Maybe see if we can give it access to steam activity or discord activity?
- The challenges may be like:
    - 2 people must talk today in a voice chat
    - 3 people must hang out in person within next week
    - Or mixed challenges (at least 1 person arts and craft, one for homework)
    - Anime Movie Night on discord
        - Auditor is encouraged to get photo to submit for VLLM to check randomly
- Evidence is posted by village member and verified via 3rd party auditor, with random checks by VLLM Agent for slop
- If lost, people's houses or levels are downgraded, or a crater is added to the village
- Rewards could be buildings created unlocking things
    - Pet Shelter
        - Unlock Pets
- Events should be scaled for rewards/downsides

Maybe every day or week have global 3rd party auditor to ensure events are properly done, with random Visual LLM verifying that the visual evidence isn't nothingness or evidence provided by auditor or else solana coin is not given

Technology Requirements:

Vultr for segment anything and 3d avatar models, on command/serverless

ElevenLabs for audio cloning for imessage messages or maybbe virtual messages on the website or both

Tiger Data for database

Auth0 for authentication, with a unique log in process

Solana for crypto currency

Stripe for payment on business side

Gemini API for llm

Avatar pipeline (No need to answer rn)

- Single-image-to-3D models (TripoSR, InstantMesh, etc.) output static meshes, not rigged characters. How will avatars walk? Options include auto-rigging, a stylized preset body with a generated head or face texture, or 2.5D billboard sprites. Pick one.
- What art style? Photoreal scans of friends look uncanny next to a stylized village. Should everything be low-poly or chibi?
- Group photo or individual photos? Group photos add a segmentation and identity-assignment step (who is which crop?).
- Is generation a one-time onboarding step, and how long can a user wait for it? Cold-start serverless GPU on Vultr can be slow.
- Which renderer: Three.js, React Three Fiber, Babylon?

The village itself

- Is the layout procedurally generated once from a seed and then persisted, or regenerated? How does it grow when someone joins or a building unlocks?
- Does the village "live" when nobody is online (server-side simulation tick), or is daily activity just generated on page load?
    - Ideally always running, use credits for Vultr and Tigerdata
- Is it multiplayer in real time (see friends' cursors or avatars live), or async only?
    - Multiplayer realtime if possible, if not thats ok
- One village per friend group, or many villages on the platform?
    - Each village is a separate instance/friend group

Voice and messaging

- iMessage has no official API. Do you mean iMessage specifically (requires a Mac relay like BlueBubbles), SMS via Twilio, Discord DMs, or in-app only?
    - Use photon and photon credits
- Consent for voice cloning and for the LLM sending messages *as* a friend. Is each person opting in? Are messages labeled as AI-generated? ElevenLabs' terms require consent from the person being cloned.
    - Voice cloning is part of onboarding
- How much voice audio per person, and is it pre-generated voice lines or live TTS per message (cost and latency)?
    - Credits from ElevenLabs
- Tone limits: is profanity and roasting on by default, and can a person set "don't roast me" or mute agent messages?
    - People can decide

Points, money, and the pot

- Clear split between XP/points (cosmetics, free) and real money (pot, competitions)? Can points be bought?
- What exactly is Stripe for? Buying cosmetics, funding the pot in fiat, or a subscription?
    - Stripe is for business end of like the people running the village, the idea is that we are a company that runs village for friend groups, and the solana is used for the rest of it
- Solana: devnet/testnet for the hackathon, or real funds? Who holds the pot, a smart contract escrow or a custodial wallet you control?
- Write out the "fancy math." Is it winner-take-all, proportional to score, or ranked tiers? What happens when only one person enters? What's the auditor's cut?
- Have you considered that pooled-money-on-outcomes can look like gambling legally?
    - This is a decentralized exchange perhaps with contracts or what not, idk solana has sponsorship, also its skill based bruh
    - For a hackathon devnet is fine, but it's worth knowing if you pitch it as a product.

Auditing and verification

- Who picks the auditor, and what stops collusion (auditor is your roommate)? Rotating, random, or voted?
    - Auditor is random every week and will sit out for the week. For competition it will be decided at competition creation so people agree with it whether auditor sits out or not, and prize pool for auditor. FOr weekly its just either based on solana purchases/bribes to LLM
- What's the grading rubric for "ambition" and "quality"? Is it the human auditor, the LLM, or both? If both, who wins on disagreement?
- What does the VLM actually check? "Image is not blank or stock," "image matches the stated goal," or "image is not AI-generated"? The last one is hard.
- Is there an appeal or dispute process?
- How are goals set so they're comparable across people (one hour of gym vs. finishing a thesis chapter)?
    - Auditor and personal taste/friend group vibes

LLM agents

- What triggers an agent message? Fixed schedule, random times, or event-driven (someone posted, someone is idle 3 days)? Max messages per person per day?
- Does the "God/Mayor" have memory across weeks, and where is that stored?
- What can a "prayer" actually do? Only pick from a vetted event list, or freeform with LLM-generated events?
- How are random events scaled for reward vs. penalty? Is there a table, or does the LLM decide?

External activity data

- Steam Web API only shows recent playtime if profiles are public. Is that acceptable?
- Discord presence needs a bot in your server with presence intent. Is everyone in one server?
- Is this read-only signal for events, or is it used as evidence for competitions?
    - Read only signal, honestly for status as well of "this person played this game for this many hours" or weekly recap

Accounts and data

- What does "unique login process" mean? Face match against your avatar photo, voice phrase, invite links, something else?
- Tiger Data is Postgres plus time-series. Which data is time-series (activity events, check-ins) versus regular relational (users, items, houses)?
- Where do photos and voice samples live, who can see them, and can someone delete themselves fully?

Cosmetics and penalties

- Who makes the predetermined assets (clothes, pets, houses)? Asset store packs, generated, hand-made? How many for the demo?
- Are penalties (house downgrade, crater) permanent or recoverable? How punishing should it feel?
    - Recoverable, but maybe with future challenge or time or shop
        - If pet dies maybe revive pet

Audit Deadlocks: If a human auditor sits out for the week, what happens if that auditor gets lazy, ghosts, or doesn't log in on Sunday night?

- Villagers have emergency vote where a majority of the village can deem a auditor “insufficient” for a week, and punish the auditor, removing a limb or killing a pet, and appointing a random auditor of those who have not been active for the week or who has been active if allowed for auditor grade themselves

Stefanny

Clothing
House Upgrades
Buildings
Village cosmetic events

Lance:

Test UI for onboarding
For evidence, creating of competition, viewing competition terms, shop ui,